

QUBO Formulation: Social Influence Maximization

Social Influence Maximization

Problem Statement. Given a social network with N candidate seed nodes, each associated with an individual influence reach w_i and a pairwise overlap w_{ij} quantifying the number of users influenced by both seed i and seed j , the objective is to select exactly K seed nodes to maximize the total unique influence coverage. The optimization must account for individual reach while penalizing double-counting of users reached by multiple selected seeds.

Decision Variables. Let $x_i \in \{0, 1\}$ for each candidate seed node $i \in \{0, 1, \dots, N-1\}$ be a binary decision variable, where $x_i = 1$ indicates that seed node i is selected, and $x_i = 0$ indicates it is not selected.

QUBO Derivation. Step 1. The original problem is a maximization. To cast it as a QUBO minimization, we negate the objective function:

$$\min_{\mathbf{x} \in \{0,1\}^N} - \sum_{i=0}^{N-1} w_i x_i + \lambda \sum_{0 \leq i < j \leq N-1} w_{ij} x_i x_j, \quad (1)$$

where $\lambda > 0$ is the penalty weight for double-counting overlap.

Step 2. The cardinality constraint requires exactly K seeds to be selected:

$$\sum_{i=0}^{N-1} x_i = K. \quad (2)$$

Step 3. We convert the constraint to a penalty term with weight $P > 0$:

$$\text{Penalty}(\mathbf{x}) = P \left(\sum_{i=0}^{N-1} x_i - K \right)^2. \quad (3)$$

Expanding the square yields:

$$\left(\sum_{i=0}^{N-1} x_i - K \right)^2 = \sum_{i=0}^{N-1} x_i^2 - 2K \sum_{i=0}^{N-1} x_i + K^2 + \sum_{i \neq j} x_i x_j. \quad (4)$$

Using the idempotent property of binary variables, $x_i^2 = x_i$, and noting that $\sum_{i \neq j} x_i x_j = 2 \sum_{0 \leq i < j \leq N-1} x_i x_j$, we obtain:

$$\left(\sum_{i=0}^{N-1} x_i - K \right)^2 = \sum_{i=0}^{N-1} (1 - 2K)x_i + 2 \sum_{0 \leq i < j \leq N-1} x_i x_j + K^2. \quad (5)$$

Multiplying by P gives the general penalty contribution to the Hamiltonian:

$$\text{Penalty}(\mathbf{x}) = P \sum_{i=0}^{N-1} (1 - 2K)x_i + 2P \sum_{0 \leq i < j \leq N-1} x_i x_j + PK^2. \quad (6)$$

Step 4. Combining the negated objective and the penalty term yields the full QUBO Hamiltonian $H(\mathbf{x})$:

$$H(\mathbf{x}) = \sum_{i=0}^{N-1} [-w_i + P(1 - 2K)] x_i + \sum_{0 \leq i < j \leq N-1} [\lambda w_{ij} + 2P] x_i x_j + PK^2. \quad (7)$$

Step 5. Reading off the coefficients from the general form $H(\mathbf{x}) = \sum_i Q_{i,i} x_i + \sum_{i < j} Q_{i,j} x_i x_j + c$, we obtain the closed-form expressions for the QUBO matrix entries and offset:

$$Q_{i,i} = -w_i + P(1 - 2K) \quad \text{for } i \in \{0, \dots, N-1\}, \quad (8)$$

$$Q_{i,j} = \lambda w_{ij} + 2P \quad \text{for } 0 \leq i < j \leq N-1, \quad (9)$$

$$Q_{j,i} = Q_{i,j}, \quad (10)$$

$$c = PK^2. \quad (11)$$

Penalty Weight Justification. Let $W_{\max} = \max_i w_i$ and $W_{\text{overlap}} = \max_{i,j} w_{ij}$. The maximum possible improvement in the original objective function achievable by violating the cardinality constraint by a single unit is bounded by $N \cdot \max_i \left(w_i + \lambda \sum_{j \neq i} w_{ij} \right)$. To guarantee that any infeasible assignment incurs a strictly higher energy than the optimal feasible assignment, the penalty weight must satisfy:

$$P > N \cdot \max_i \left(w_i + \lambda \sum_{j \neq i} w_{ij} \right). \quad (12)$$

This condition ensures that the energy cost of constraint violation dominates any potential gain in the objective function, preserving the equivalence between the constrained optimization problem and the unconstrained QUBO minimization.

Correctness Argument. Minimizing $H(\mathbf{x})$ over $\{0, 1\}^N$ recovers the optimal solution because (i) for any feasible assignment satisfying $\sum_i x_i = K$, the penalty term vanishes identically, and $H(\mathbf{x})$ reduces exactly to the negated original objective; thus, minimizing

H over the feasible subspace is equivalent to maximizing the original influence coverage. (ii) For any infeasible assignment, the penalty term contributes a strictly positive energy $P(\sum_i x_i - K)^2$. By selecting P according to the bound derived above, the energy of every infeasible point strictly exceeds that of the optimal feasible point. Consequently, the global minimum of $H(\mathbf{x})$ must occur at a feasible assignment that maximizes the original objective, ensuring exact recovery of the optimal seed set.

Illustrative Example. Consider a concrete instance with $N = 3$ candidates, $K = 2$, influence reaches $w_0 = 10$, $w_1 = 8$, $w_2 = 9$, pairwise overlaps $w_{01} = 3$, $w_{02} = 2$, $w_{12} = 4$, overlap penalty $\lambda = 1$, and constraint penalty $P = 100$. Substituting these values into the general formulas yields:

$$Q_{0,0} = -10 + 100(1 - 4) = -310, \quad (13)$$

$$Q_{1,1} = -8 + 100(1 - 4) = -298, \quad (14)$$

$$Q_{2,2} = -9 + 100(1 - 4) = -309, \quad (15)$$

$$Q_{0,1} = 1 \cdot 3 + 200 = 203, \quad (16)$$

$$Q_{0,2} = 1 \cdot 2 + 200 = 202, \quad (17)$$

$$Q_{1,2} = 1 \cdot 4 + 200 = 204, \quad (18)$$

$$c = 100 \cdot 2^2 = 400. \quad (19)$$

The resulting Hamiltonian is:

$$H(x_0, x_1, x_2) = -310x_0 - 298x_1 - 309x_2 + 203x_0x_1 + 202x_0x_2 + 204x_1x_2 + 400. \quad (20)$$

Enumerating the three feasible solutions ($\sum_i x_i = 2$):

$$\mathbf{x} = (1, 1, 0) \implies H = -310 - 298 + 203 + 400 = -5, \quad (21)$$

$$\mathbf{x} = (1, 0, 1) \implies H = -310 - 309 + 202 + 400 = -17, \quad (22)$$

$$\mathbf{x} = (0, 1, 1) \implies H = -298 - 309 + 204 + 400 = -3. \quad (23)$$

The global minimum is $H^* = -17$ at $\mathbf{x}^* = (1, 0, 1)$, corresponding to selecting seed nodes 0 and 2. The maximum influence coverage is $10 + 9 - 1 \cdot 2 = 17$, matching the negated optimal energy offset by the constant c .